

Investigating the effect of Input Modality on Reasoning in VLMs using Zero-shot evaluation

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Abstract

This project investigates how input modality affects mathematical reasoning in vision-language models (VLMs). Using 100 samples from the GSM8K benchmark [1], we evaluate Qwen2-VL-2B-Instruct [2] and LLaVA-v1.6-Mistral-7B [3] across three conditions: text-only, rendered image, and a modality mismatch condition where image and text describe different problems. Both models show substantial accuracy drops under visual input — Qwen2 falls from 55% to 30% and LLaVA from 40% to 21%. In the mismatch condition, both models follow the text modality in over 75% of cases, revealing strong text dominance when modalities conflict. These findings suggest that rendering math problems as images meaningfully impairs reasoning in current VLMs, even when visual content is clean and legible.

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References

- [1] Cobbe et al. (2021). *Training Verifiers to Solve Math Word Problems*. arXiv:2110.14168.
- [2] Wang et al. (2024). *Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution*. arXiv:2409.12191.
- [3] Liu et al. (2024). *LLaVA-NeXT: Improved Baselines with Visual Instruction Tuning*. arXiv:2310.03744.